

Type 2 Diabetes Digital Twin

Forecast glucose 2 hours ahead. Warn early. Test remedies.

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Happiest Health Digital Twin Challenge 2026

github.com/venkatesh-spec01/diabetes-digital-twin

Research prototype on 100% synthetic data. Not medical advice.

The problem

- People with Type 2 Diabetes often learn about a glucose spike only after it has happened.
- A twin that updates continuously can warn earlier, and show which small change helps most.
- Condition: Type 2 Diabetes.
- Adverse events predicted: glucose above 180 mg/dL (HIGH) and below 70 mg/dL (LOW), up to 2 hours ahead.
- Small changes tested: a 10-minute walk, a 30% smaller portion, a low-GI swap.

The solution: a six-layer twin

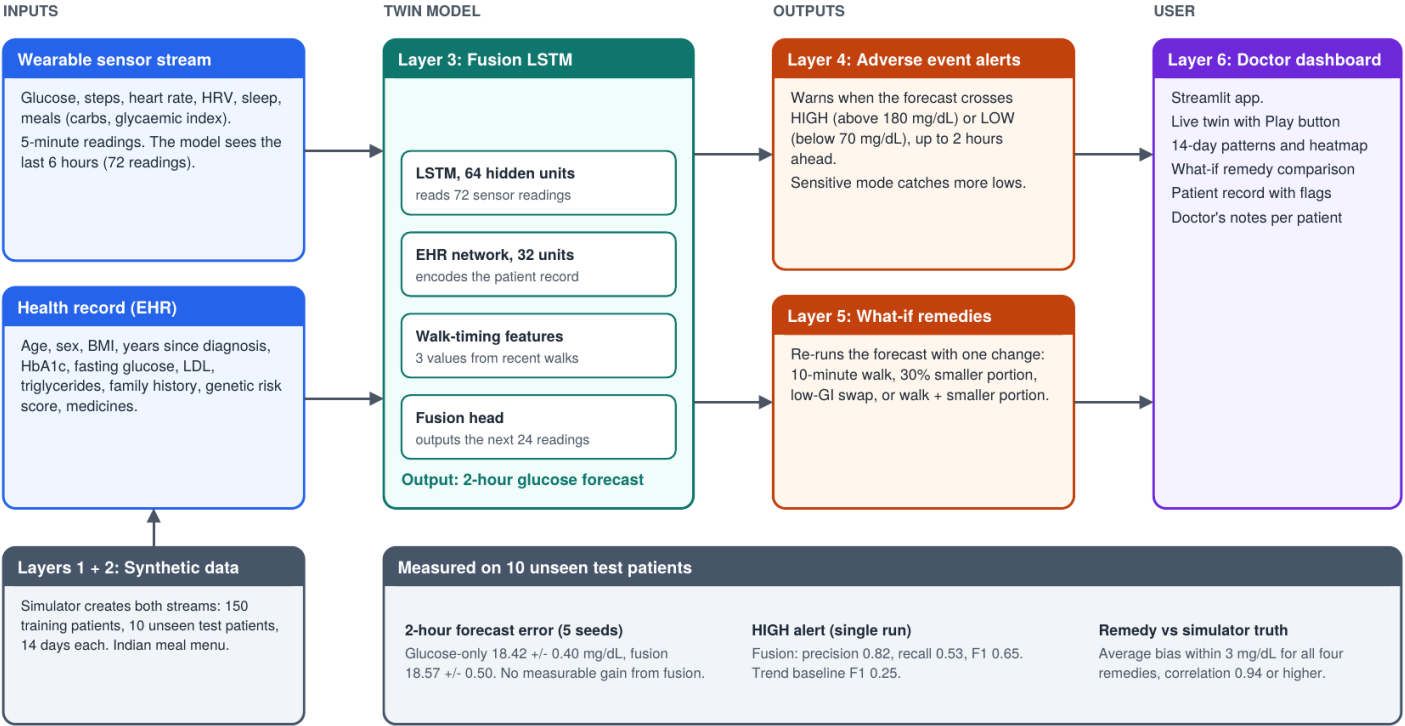
Layer	What it does
1 + 2	Synthetic EHR + wearable generator (glucose, steps, HR, HRV, sleep, meals)
3	Fusion LSTM: 6 h of sensor history + patient record, predicts next 2 h
4	Alerts for HIGH (>180) and LOW (<70) with precision / recall / F1
5	What-if remedies, checked against the simulator's true answers
6	Streamlit doctor dashboard
Check	5-seed reproducibility test

Stack: Python 3.14.8, PyTorch 2.14.1, Streamlit, Plotly. Settings: 72 readings history, 24 readings horizon, hidden 64, 12 epochs.

Architecture

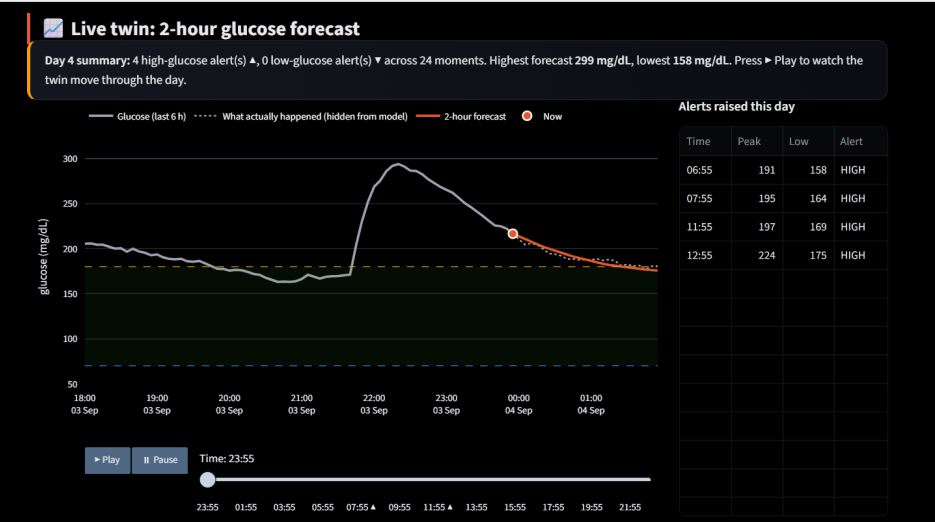
Type 2 Diabetes Digital Twin: System Architecture

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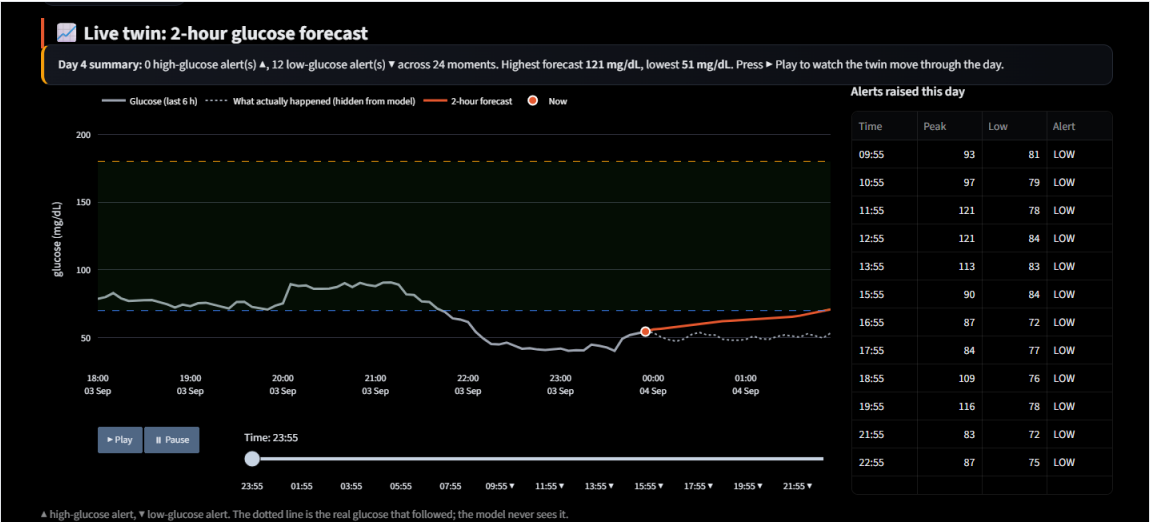


Research prototype on 100% synthetic data. Not medical advice and not validated on real patients.
Code: github.com/venkatesh-spec01/diabetes-digital-twin | Python 3.14.8, PyTorch 2.14.1, Streamlit

Doctor dashboard: live twin and alerts



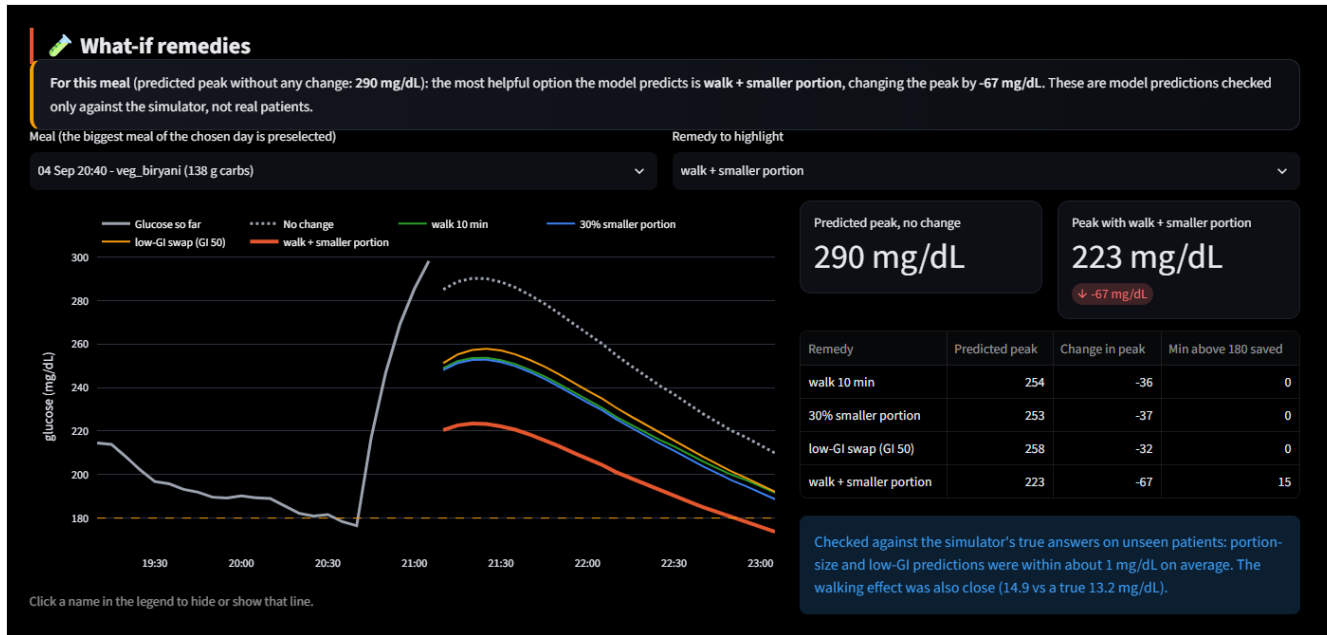
HIGH alert (T03, day 4: 4 alerts)



LOW alert (T06, day 4: 12 alerts)

- Play button replays the day; forecast appears 2 hours ahead of the real curve.
- Sensitive mode is the default because a missed low is the more dangerous error.

What-if remedies and 14-day patterns



- Pick a meal, compare remedies side by side.
- T03 shown with a 138 g-carb meal, an extreme case. Average walk + smaller portion effect is about 20 mg/dL.
- T03 over 14 days: 56% in range, 44% above 180.



Results on 10 unseen patients

2-hour forecast error (RMSE, 5 seeds)

Model	RMSE mg/dL
Glucose-only LSTM	18.42 +/- 0.40
Fusion LSTM	18.57 +/- 0.50

HIGH alerts: 1,661 events, 10,695 windows

Method	Precision	Recall	F1
Trend baseline	0.37	0.19	0.25
Glucose-only LSTM	0.76	0.49	0.59
Fusion LSTM	0.82	0.53	0.65
Fusion, sensitive	0.54	0.85	0.66

LOW alerts: 239 events

Method	Prec.	Recall	F1
Fusion, default	1.00	0.14	0.24
Fusion, sensitive	0.40	0.82	0.53

- Fusion is not a measurable RMSE gain (better in 2 of 5 seeds). We do not claim one.
- Alerts: fusion beats the trend baseline 0.65 vs 0.25 F1.
- Default setting: 0 false low alarms but misses most lows. Sensitive catches 82%.
- Alert numbers are a single run.

Remedies vs the simulator's true answers

Remedy	True drop	Predicted	Bias	MAE	Corr.
Walk 10 min	13.23	14.90	+1.67	2.37	0.96
30% smaller portion	11.29	11.42	+0.13	2.60	0.94
Low-GI swap (GI 50)	11.97	11.91	-0.06	2.59	0.94
Walk + smaller portion	20.42	23.12	+2.70	3.74	0.96

- Average bias within 3 mg/dL for all four remedies; correlation 0.94 or higher.
- Walk is slightly overestimated; the combined remedy is slightly optimistic (+2.7 mg/dL).
- Checked against the same simulator that made the data: this tests internal consistency, not real physiology.

Limitations and next steps

Limitations

- All data is synthetic; never seen a real patient.
- Remedy checks use the same simulator.
- Fusion gives no 2-hour RMSE gain.
- Walk remedy slightly optimistic.
- Alert numbers from a single run.
- Not a medical device.

Validation plan

- Test on a public real CGM dataset; report the gap.
- Add prediction intervals and Clarke Error Grid.
- Tune thresholds on a separate validation set.
- Report alerts per episode and warning time.
- Clinician pilot under ethics approval.

Outcome

- Working end-to-end twin: data, forecasting, alerts, remedies, dashboard.
- Fusion LSTM raises HIGH-alert F1 from 0.25 (trend) to 0.65.
- Remedy predictions track the simulator's truth within 3 mg/dL on average.
- Honest reporting: where fusion does not help, we say so.
- Open source (MIT). Reproducible: pinned requirements, seeds, committed data and models.
- Code: github.com/venkatesh-spec01/diabetes-digital-twin